# Assignment 2: RFE


**Summary**

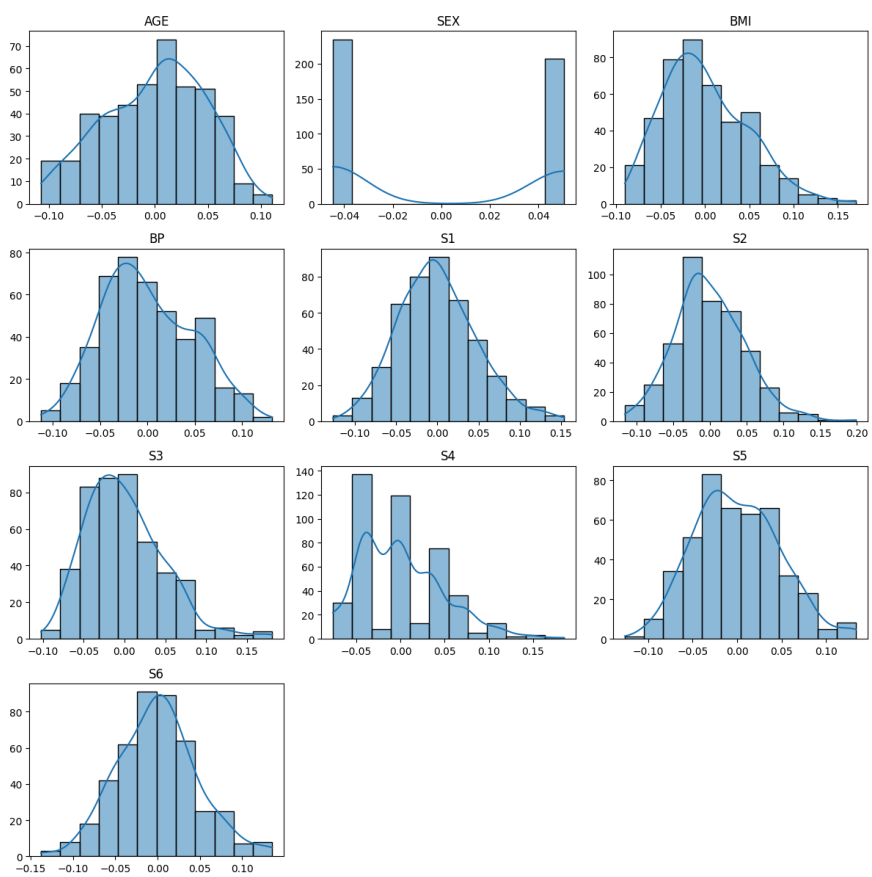
This assignment uses the Diabetes dataset to predict the outcome variable. I began by exploring the dataset and building a baseline Linear Regression model. Next, I applied Recursive Feature Elimination (RFE) to remove features step by step while monitoring changes in the $R^2$ score. Finally, I examined which features were most important and compared the initial coefficient-based ranking with the features selected by RFE


**Dataset Exploration**

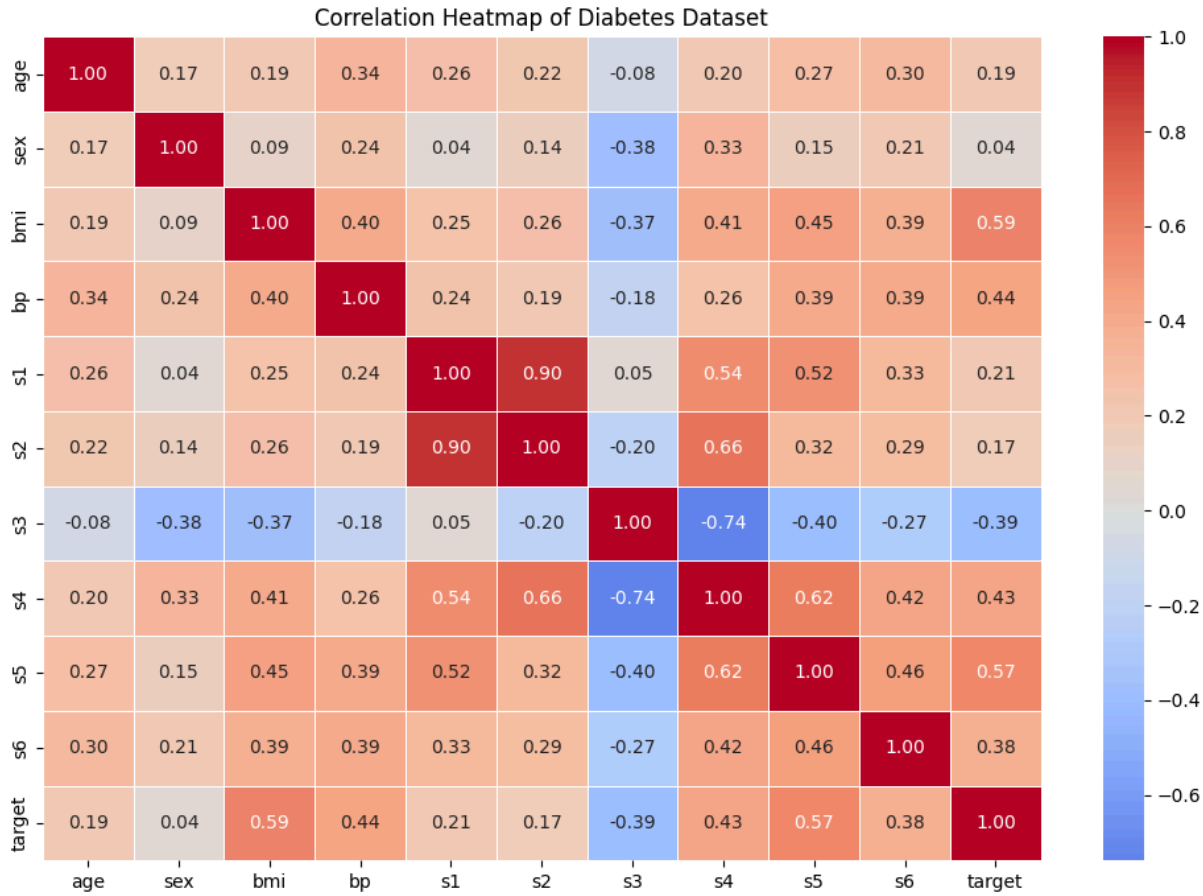
The dataset contains 442 observations and 10 predictor variables, along with one target variable. All 11 columns are numeric (float type), and there are no missing values, meaning the dataset is already clean and ready for analysis.

Based on the summary statistics, the input features appear to be standardized — their means are close to 0 and their standard deviations are very similar.

In contrast, the target variable shows a wide range of values, from approximately 25 to 346, with an average of about 152.




Most features are roughly bell-shaped and centered around 0, consistent with standardization. **SEX** has only two values, so it appears as two spikes, and features like **BMI** and **S4** show slight skewness rather than a perfectly normal shape.

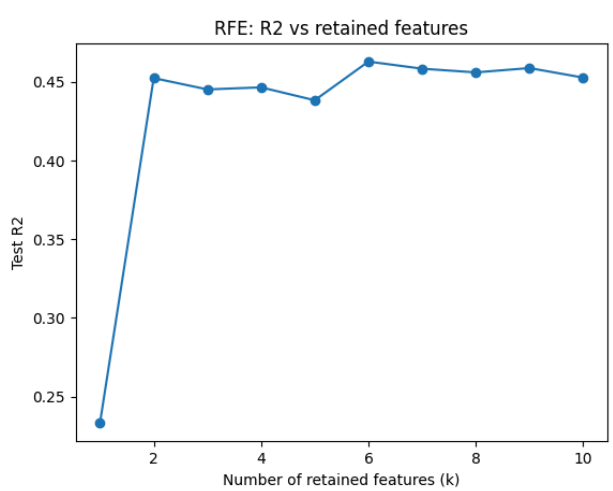



The target variable has the strongest positive correlation with **bmi (~0.59)** and **s5 (~0.57)**, while **bp** and **s4** show moderate positive correlations. Some predictors are also highly correlated with each other, for example, **s1 and s2 (~0.90)** are strongly positively related, whereas **s3 and s4 (~−0.74)** have a strong negative relationship.

**<u>Model and Feature Selection Findings:</u>**

1. Baseline model (Linear Regression)
   After training a Linear Regression model on the training data, the test $R^2$ was about 0.45. This means the model explains roughly 45% of the variation in the target using all 10 features.

2. RFE Plot ($R^2$ vs. Number of Retained Features)
   The $R^2$ increases significantly when moving from 1 to 2 features, then remains mostly stable afterward. Since the improvement beyond k = 2 is less than 0.01, choosing k = 2 is reasonable because it keeps the model simpler without sacrificing much accuracy.



3. Feature Ranking (Initial Coefficients vs. RFE Ranking)
   In the baseline model, the largest absolute coefficients were for s1, s5, and bmi, indicating they had the strongest influence on predictions. After applying RFE with k = 2, the selected features were bmi and s5, which align with the top baseline rankings. The model likely removed s1 because it overlaps with related features and adds limited additional predictive value.

```
Initial ranking by coef:
 s1     931.488846
s5     736.198859
bmi    542.428759
s2     518.062277
bp     347.703844
s4     275.317902
sex    241.964362
s3     163.419983
s6      48.670657
age     37.904021
dtype: float64

Top 3: ['s1', 's5', 'bmi']

Final selected features (k=2): ['bmi', 's5']
Selected positions in initial ranking: {'bmi': 3, 's5': 2}

RFE ranking (1=best):
 bmi     1
s5      2
s1      3
s2      4
bp      5
sex     6
s4      7
s3      8
s6      9
age    10
dtype: int64
```

**Conclusion**

Overall, the Linear Regression model provides a reasonable baseline performance on this dataset, explaining about 45% of the variation in the target. The RFE results show that nearly the same level of accuracy can be achieved using only two features, demonstrating that the model does not require all predictors to perform well.

The selected features (bmi, s5) appear to contain most of the meaningful predictive information. This indicates that while the dataset includes several variables, only a few contribute strongly to the prediction, and adding more features beyond the top ones leads to only minor improvements in performance.